Football Match Outcome Prediction Using Machine Learning

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1. Introduction

Football match prediction is a common machine learning problem where historical match data is used to estimate the result of future matches. In this project, we created a simple machine learning model that predicts the outcome of an international football match.

The model uses past match information such as home team, away team, tournament, country, neutral venue, year, and final score. Based on these historical records, the model learns patterns and predicts the expected goal difference between two teams.

The predicted goal difference is then used to decide whether the home team wins, the away team wins, or the match ends in a draw.

2. Project Objective

The main objective of this project is to predict the result of a football match using machine learning.

The model predicts:

- Home team win
- Away team win
- Draw

Instead of directly predicting the winner, the model first predicts the goal difference:

goal_difference = home_score - away_score

Then the goal difference is converted into a match result.

3. Dataset Used

The dataset used in this project was downloaded from Kaggle. The downloaded dataset contained multiple files, but in this project, we used only:

results.csv

This file was used because it contains the most important match-level information needed for prediction.

The important columns in the dataset were:

date

home_team

away_team

home_score

away_score

tournament

country

neutral

The other files, such as shootouts.csv, goalscorers.csv, and former_names.csv, were not used in this simplified version.

4. Tools and Libraries Used

This project was completed using Anaconda Jupyter Notebook.

The main Python libraries used were:

pandas

numpy

matplotlib

scikit-learn

pandas was used for loading and handling the dataset.

numpy was used for numerical operations.

matplotlib was used for visualization.

scikit-learn was used for building and evaluating the machine learning model.

5. Loading the Dataset

The dataset was loaded into Jupyter Notebook using pandas.

```
results = pd.read_csv(r"C:\Users\ajeet\Downloads\archive\results.csv")
```

After loading the dataset, the first few rows were checked using:

```
results.head()
```

This helped confirm that the dataset was imported correctly.

6. Data Preparation

After loading the dataset, we created a copy of the original data.

The date column was converted into proper datetime format, and a new column called year was created.

```
data["date"] = pd.to_datetime(data["date"])

data["year"] = data["date"].dt.year
```

Rows with missing scores were removed because the model needs completed match results for training.

This step helped clean the data before applying machine learning.

7. Creating the Target Variable

The target variable is the value that the machine learning model tries to predict.

In this project, the target variable was:

goal_difference

It was calculated using:

goal_difference = home_score - away_score

For example:

Brazil 4 - 1 South Korea

goal_difference = 4 - 1 = 3

A positive goal difference means the home team performed better.

A negative goal difference means the away team performed better.

A goal difference close to zero means the match is likely to be a draw.

8. Feature Selection

The input features used for prediction were:

year

home_team

away_team

tournament

country

neutral

These features were selected because they are available before a match is played.

The columns home_score and away_score were not used as input features because they are the final result. Using them as input would be incorrect because the model would already know the answer.

9. Encoding Categorical Data

Some columns in the dataset contained text values, such as team names and tournament names.

Machine learning models cannot directly understand text, so these columns were converted into numbers using LabelEncoder.

The encoded columns were:

home_team

away_team

tournament

country

neutral

This allowed the model to process the data properly.

10. Splitting the Data

The dataset was split into training and testing data.

The training data was used to teach the model.

The testing data was used to check how well the model performs on unseen data.

We used an 80-20 split:

80% training data

20% testing data

This was done using train_test_split.

11. Machine Learning Model Used

In this project, we used a Linear Regression model.

Linear Regression is a supervised machine learning algorithm that predicts a numerical value.

Here, the numerical value predicted by the model was:

goal_difference

The model learned patterns from historical football matches and used those patterns to predict the expected goal difference for a new match.

12. Model Training

The Linear Regression model was trained using the training data.

linear_model = LinearRegression()

linear_model.fit(X_train, y_train)

During training, the model learned the relationship between the input features and the goal difference.

13. Model Evaluation

After training, the model was tested using the testing data.

The model predicted goal differences for matches it had not seen before.

The predicted values were compared with the actual goal differences.

The evaluation metrics used were:

Mean Absolute Error

Mean Squared Error

Mean Absolute Error shows the average prediction error.

Mean Squared Error gives more penalty to larger errors.

These metrics helped us understand how close the model's predictions were to the actual results.

14. Making a Prediction

A prediction function was created to predict a new match.

The function takes details such as:

home team

away team

tournament

country

neutral venue

year

Then the model predicts the expected goal difference.

The result is decided like this:

If predicted goal difference > 0.3:

Home team wins

If predicted goal difference < -0.3:

Away team wins

If predicted goal difference is between -0.3 and 0.3:

Draw

Example:

predict_match_linear("Argentina", "France")

This gives the predicted goal difference and the predicted match result.

15. How the Prediction Works

The prediction is made by using historical football match data.

First, the model studies previous matches and learns how different teams performed in different situations.

It looks at information such as:

Which team was home

Which team was away

Which tournament was played

Where the match was played

Whether the match was neutral

What year the match happened

What the final score was

From the final score, the model learns the goal difference.

When we give the model two countries for a new match, it uses the historical patterns to predict the expected goal difference.

Based on that predicted goal difference, the model decides the match result.

16. Limitations

This is a simple machine learning project, so it has some limitations.

The model does not include advanced football factors such as:

recent team form

FIFA ranking

Elo rating

player injuries

team lineups

manager tactics

penalty history

weather

red cards

current squad strength

Also, team names were encoded as numbers, which is simple but does not fully represent the actual strength of each team.

Because of these limitations, the result should be considered a basic prediction, not a guaranteed outcome.

17. Conclusion

In this project, we built a simple machine learning model to predict football match outcomes.

We used historical international football match data from results.csv. The model used match information such as home team, away team, tournament, country, neutral venue, and year.

The model predicted the expected goal difference between two teams. Based on the predicted goal difference, the match was classified as a home win, away win, or draw.

This project demonstrates a basic machine learning workflow using Python, Jupyter Notebook, and scikit-learn.